HearEdit

Speaker Diarization System

Milestone 1 · Sprint 1 & 2

Agenda

Milestone 1

01 Project Overview

02 Use Case & Dataset

03 EDA

04 Model Approach

05 Results & Threshold

Project Overview

Project Name

HearEdit
Speaker Diarization System

Use Case

Identify & separate speakers
in multi-speaker audio

Real-world Application

Meeting transcription
Call center analytics
Podcast indexing

Dataset & Exploratory Data Analysis

LibriSpeech (dev-clean)

- Open-source English audiobook dataset
- Multiple speakers, multiple chapters each
- Audio in .flac format, converted to .wav at 16kHz
- Paired with transcription files (.trans.txt)
- Speaker metadata: ID, gender, subset, name

EDA Findings

Outlier removal

IQR method on #samples per speaker (1.5×IQR)

Data balance

Speakers with too few or too many samples removed

Embeddings

128-dim vectors extracted via Vosk SpkModel

Normalization

L2-normalized embeddings before similarity computation

Model Approach

1 — 2 — 3 — 4 — 5

Audio Input

Raw .flac audio → converted to .wav (16kHz mono) via ffmpeg

Feature Extraction

Vosk SpkModel extracts 128-dim voice embeddings per segment

Cosine Similarity

Each embedding vs. known speaker prototypes (pairwise)

Threshold Decision

EER threshold decides: known speaker, best match, or new speaker

Prototype Update

Running mean of all vectors per speaker is maintained

💡 Alternatives considered: one-shot neural network, GMM — Cosine similarity chosen for simplicity, speed & reliability

Model Evaluation & Threshold Selection

EER

Criterion

Equal Error Rate
balances FAR & FRR

ROC

Curve

AUC computed for
threshold selection

~0.4370

EER Threshold

Cosine similarity
cutoff value

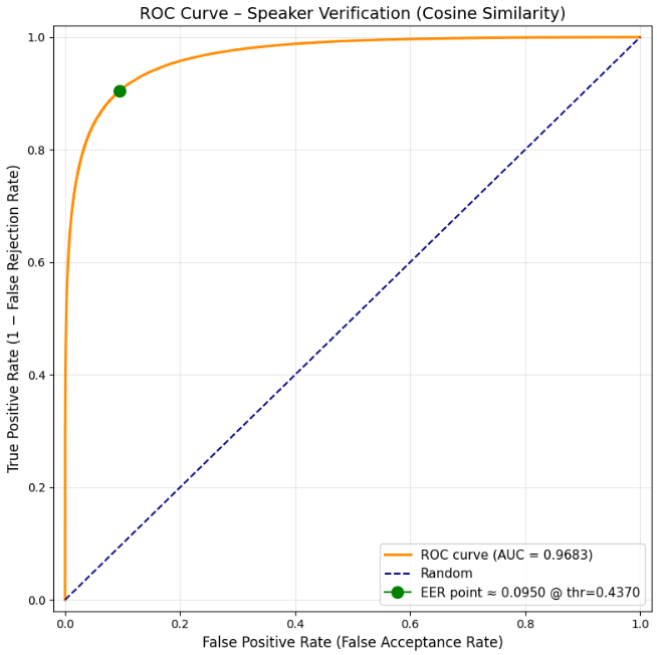
Threshold Methodology

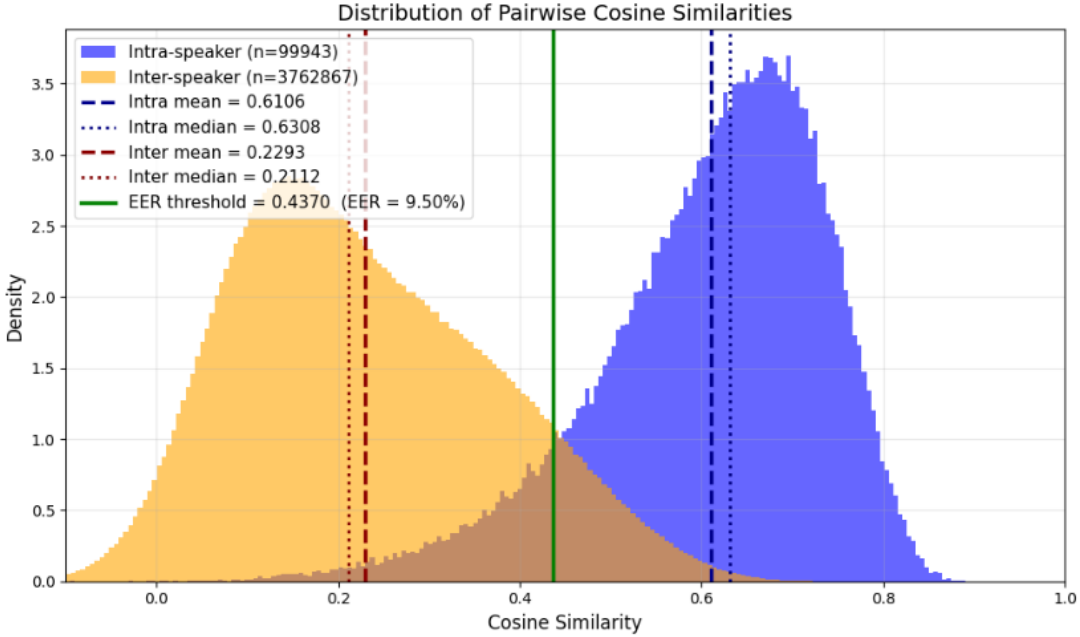
- Pairwise intra-speaker similarities computed
- Pairwise inter-speaker similarities computed
- EER: point where FAR = FRR on ROC curve
- Avoids bias toward one error type
- Result: balanced speaker separation

Distribution Observations

- Intra-speaker similarities: higher & tighter cluster
- Inter-speaker similarities: lower, broader spread
- Some overlap → threshold needed for disambiguation
- Mean intra > mean inter → discriminability confirmed

Graphical Analysis





Thank you

Questions?

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